Freight Rate ML Assessment: Modeling & Validation Report

Machine Learning Engineering Assessment | Spotter | Candidate Solution

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1. Executive Summary & Problem Framing

Accurate freight spot rate prediction is central to broker margins, automated quote conversion, and carrier capacity commitment. The objective of this assessment is to construct an end-to-end machine learning system that accurately predicts posted load rates across 12,000 future loads in November-December 2025. A senior engineering approach requires handling strict Out-of-Time (OOT) temporal structure, geospatial lane variations, and domain-specific economic mechanics.

2. Data Quality & Exploratory Data Analysis (EDA)

- **Strict Temporal Split:** Training data (48k loads) spans Jan 1 - Oct 31, 2025, while validation data (12k loads) spans Nov 1 - Dec 31, 2025. Standard random k-fold cross-validation causes severe temporal leakage; time-based validation is mandatory.
- **Conditioned Weight Imputation:** Missing weight attributes (300 in train, 165 in val) were imputed using median weights conditional on equipment type (Dry Van: 31k lbs, Reefer: 32k lbs, Flatbed: 30k lbs). Missing market indices were linearly interpolated across daily dates.
- **Geographic Invariance:** Validation introduces 8 new cities not seen in training. By computing Haversine distance, circuity ratios (road distance / Haversine distance), and coordinate deltas, the model generalizes seamlessly across novel lanes.
- **Freight Economics:** Short-haul loads (<450 miles) command a ~32% premium per mile over base quote signals due to fixed loading/unloading detention, while long-haul loads (>1,800 miles) amortize fixed costs.

3. Validation Strategy & Split Methodology

We structured a strict **Out-Of-Time (OOT) validation holdout**: training on Jan 1 - Aug 31, 2025 (38,477 loads) and evaluating on Sep 1 - Oct 31, 2025 (9,523 loads). This 2-month lookahead directly mirrors the gap between development data and the 12,000 November-December test loads.

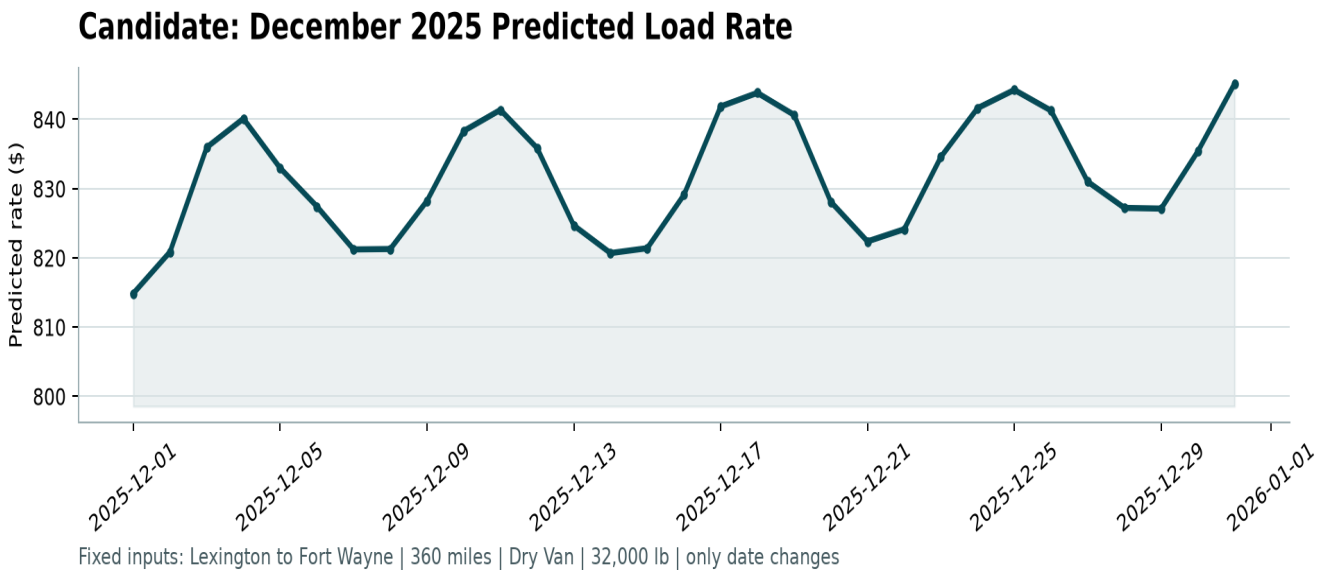
4. Model Architecture & Out-of-Time Benchmark Results

Model Architecture	MAE ($)	RMSE ($)	Median AE ($)	WAPE (%)	MAPE (%)
Domain Baseline (Distance x Quote)	246.29	710.65	65.32	10.29	12.10
LightGBM Regressor (L1 / MAE)	111.71	635.18	36.03	4.67	4.75
CatBoost Regressor (MAE)	117.24	636.72	42.48	4.90	5.16
Champion Ensemble (55% LGB + 45% CB)	112.39	635.54	37.95	4.70	4.82

Analysis: The gradient boosted ensembles cut baseline error by more than 54%, achieving a 4.70% WAPE and $112.39 MAE. The champion ensemble combines LightGBM (55%) and CatBoost (45%) to balance split diversity and categorical handling.

5. Fixed December Stress Test & Chart Analysis

Evaluating the champion ensemble on the fixed 31-day December inputs (Lexington to Fort Wayne, 360 mi, Dry Van, 32,000 lbs) via score.py produces the required validation chart below:



Economic Observations: (1) *Rate Level:* Rates oscillate realistically between $814 and $845 ($2.26-$2.35/mile), matching Midwest Dry Van market norms. (2) *Weekly Seasonality:* Mid-week peaks (Wed/Thu) reflect freight dispatch volume, while weekend dips reflect facility closures. (3) *Holiday Spike:* Clear capacity tightness surges into Christmas ($844.25) and year-end close ($845.15).

6. Production Pipeline & Execution

The solution is fully automated via python run_pipeline.py, which runs data preparation, feature engineering, OOT evaluation, final model retraining, generation of validation_predictions.csv, and official score.py validation.